MLOps Maturity Ladder

Ad hoc to fully automated pipelines

PROFESSIONAL DEFINITION

The MLOps Maturity Ladder describes stages of operational capability for machine learning, from ad hoc manual work to fully automated, self-healing pipelines, indicating the cost and reliability at each stage.

WHO DEVELOPED IT

Formalised in industry MLOps guidance, including Google Cloud's MLOps maturity levels and Microsoft's MLOps practices.

WHY IT WAS DEVELOPED

It was developed to help organisations locate their ML operational maturity and the investment required to scale models reliably.

FIGURE 1 · MLOPS MATURITY LADDER

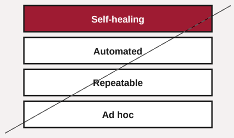

Figure 1. A schematic of the MLOps Maturity Ladder as applied in BARBM312 (illustrative).

HOW IT IS USED

Current practice is placed on a rung; the next rung defines the automation, monitoring and governance investment needed to scale.

WHEN IT IS USED

When scaling AI from pilot to production and planning MLOps investment.

BY WHOM IT IS USED

ML engineering, platform and operations leaders.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

It turns reliable ML operations into a staged capability, exposing the often-underestimated cost of moving beyond prototypes.

RELATED FRAMEWORKS IN THIS COURSE

- **Open vs. Proprietary LLM 2×2** · Slibrary 11
- **Make-vs-Buy Decision Tree** · Slibrary 11
- **Talent Scarcity Heatmap** · Slibrary 11
- **Skill-Premium Curve** · Slibrary 11

SOURCES (HARVARD REFERENCING STYLE)

Google Cloud (2020) 'MLOps: Continuous delivery and automation pipelines in machine learning'.

Kreuzberger, D., Kühl, N. and Hirschl, S. (2023) 'Machine learning operations (MLOps)', IEEE Access, 11.